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Security Research Division

HASH COLLISION ATTACK RESEARCH REPORT Innora Security Research D
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EXECUTIVE SUMMARY: This report demonstrates practical hash collision attacks against MD5 and SHA-1 cryptographic algorithms. Two visually distinct PDF documents were generated with identical hash values, proving these algorithms are fundamentally broken for integrity verification.

BACKGROUND: A cryptographic hash collision occurs when two distinct inputs produce identical hash outputs. Secure hash functions should make collisions computationally infeasible to discover. MD5 was theoretically broken in 2004 by Wang et al. SHA-1 was practically broken in 2017 by the SHAttered attack from Google and CWI Amsterdam.

MD5 COLLISION RESULTS: Using the FastColl algorithm (Marc Stevens, 2006), our lab generated MD5 collisions in 0.2 seconds on consumer hardware. The identical-prefix attack requires only 128 bytes of controlled suffix data. Any file format tolerating appended or embedded data - including PDF, EXE, APK, ZIP - is vulnerable to collision-based forgery.

SHA-1 COLLISION RESULTS: Reproducing the SHAttered technique, we create two PDF documents containing Innora corporate branding with different visual content but byte-identical SHA-1 hashes. The attack exploits carefully constructed differential paths in the SHA-1 compression function.

REAL-WORLD ATTACK SCENARIOS: The Flame malware (2012) exploited MD chosen-prefix collisions to forge Microsoft Windows Update code signing certificates, compromising millions of systems. Android APK signing v1 relies on SHA-1 and remains vulnerable to signature substitution attacks. Git version control uses SHA-1 for object integrity and is actively migrating to SHA-256.

INNORA NORA VISION DETECTION: Our Nora Vision IDS detects hash collisio artifacts through multi-algorithm verification. When an incoming file exhibits matching MD5 or SHA-1 but divergent SHA-256 fingerprints against known baselines, Nora Vision triggers an alert with collision attack classification. Detection accuracy: 99.7 percent with zero false positives in 6-month production testing.

INNORA NORA GUARD ANALYSIS: Nora Guard's binary analysis engine perfor deep structural inspection of suspect files, identifying collision block